Skills Summary

Choosing Linear, Quadratic, or Piecewise Models

Use this reference while completing your worksheet or when studying.

The model is a *conclusion*, not a starting assumption. Read the evidence — a difference table, or the wording of the story — decide the family, and only then write a formula.

1. Evidence for a linear model

Evidence: equal steps in the input give **constant first differences** in the output; or the story names one fixed rate that never changes ("$15 per hour", "5 cm each minute").

Model: $y = mx + b$, where m is the constant difference per unit step and b is the output when $x = 0$. If the table does not start at $x = 0$, find b by substituting any one row.

(!) A start-up fee is added *once*. Doubling the input never doubles the output unless $b = 0$.

Example: For $x = 0, 1, 2, 3$ a table gives $y = 3, 7, 11, 15$. Predict y at $x = 9$.
Step 1. The first differences are $4, 4, 4$ — constant — so the family is linear and I only need a slope and an intercept.
Step 2. $m = 4$ and $y = 3$ at $x = 0$, so $y = 4x + 3$, and at $x = 9$ that is $4(9) + 3$.
39

Try it: For $x = 0, 1, 2, 3$ a table gives $y = 2, 5, 8, 11$. Predict y at $x = 8$.

check: **26**

2. Evidence for a quadratic model

Evidence: the first differences *change*, but they change by the same amount each time — **constant second differences**. In a story: anything that multiplies two quantities which both depend on the input (area, revenue = price × quantity, projectile height).

Model: $y = ax^2 + bx + c$. The second difference equals $2a$ for unit steps, so a second difference of -32 gives $a = -16$.

(!) Never extend a quadratic table with its first difference. That treats a changing rate as constant and the error grows with every step.

Example: For $x = 0, 1, 2, 3$ a table gives $y = 1, 4, 11, 22$. Predict y at $x = 5$.
Step 1. First differences $3, 7, 11$ are not constant but the second differences are both 4, so the family is quadratic, not linear.
Step 2. Second difference 4 gives $a = 2$; $c = 1$ from $x = 0$; and $x = 1$ gives $2 + b + 1 = 4$, so $b = 1$. Then $y = 2x^2 + x + 1$ at $x = 5$ is $50 + 5 + 1$.
56

Try it: For $x = 0, 1, 2, 3$ a table gives $y = 2, 6, 16, 32$. Predict y at $x = 4$.

check: **54**

3. Evidence for a piecewise model

Evidence: the story states a boundary where the *rule* changes — "the first 3 days… after that…", "up to 6 miles… beyond 6 miles…". One formula cannot describe both sides.

$$F(t) = \begin{cases} \text{flat fee}, & t \le \text{boundary} \\ \text{flat fee} + \text{rate} \times (t - \text{boundary}), & t > \text{boundary} \end{cases}$$

(!) Subtract the boundary before multiplying, and after solving check that your answer really lies in the piece you used.

Example: A locker costs 5 dollars for the first 3 days and 2 dollars for each day after that. Find the cost of 7 days.
Step 1. The rule changes at day 3, so this is piecewise and only the days past the boundary are charged at the daily rate.
Step 2. Days past the boundary: $7 - 3 = 4$, so the cost is $5 + 2(4)$.
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Try it: A rental costs 6 dollars for the first 4 hours and 3 dollars for each hour after that. Find the cost of 9 hours.

check: **21**

4. Comparing two models

"When do the two agree?" means **set the models equal and solve**, then substitute back to get the matching output.

Two lines meet at most once (or never, if the slopes match). **A line and a parabola** can meet twice, once, or not at all — so finding two answers is normal, not an error.

Read the meeting point: it is where the cheaper option changes. Before it one model wins; after it the other does.

Example: Find every point where $y = x^2 + 2$ and $y = 3x$ agree.
Step 1. Agreement means equal outputs at the same input, so I set the two expressions equal — and because one is quadratic I expect up to two answers.
Step 2. $x^2 + 2 = 3x$ gives $x^2 - 3x + 2 = 0$, so $(x-1)(x-2) = 0$ and $x = 1$ or $x = 2$; then $y = 3x$ gives 3 and 6.
(1, 3) and **(2, 6)**

Try it: Find the point where $y = 5 + 3x$ and $y = 11 + x$ agree.

check: **(3, 14)**